

CURELIO

AI Powered Cashless Health Insurance Eligibility Prediction System



PROJECT REPORT

Python | Machine Learning | Flask

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ABSTRACT

Curelio is a Machine Learning based Cashless Health Insurance Eligibility Prediction System designed to simplify the insurance verification process between patients, hospitals, and insurance providers.

Traditional cashless insurance approval requires manual verification, which can increase waiting time and create delays for patients. Curelio uses machine learning techniques to analyze healthcare-related information such as insurance provider, hospital details, policy type, and treatment information to predict cashless eligibility.

The system is developed using Python and Flask, with machine learning models trained and compared to identify the most suitable algorithm. After evaluating multiple algorithms, the best performing model is integrated into a web application to provide fast and user-friendly eligibility predictions.

This project demonstrates how Artificial Intelligence and Machine Learning can support healthcare systems by improving decision-making, reducing manual effort, and providing quicker assistance during the insurance verification process.

Keywords: Machine Learning , Healthcare, Insurance Eligibility, Flask, Decision Tree Algorithm

ACKNOWLEDGMENT

We would like to express our sincere gratitude to DEVSPRINT for providing us with the opportunity to work on this internship project and gain practical experience in developing real-world technology solutions.

We are thankful for the guidance, support, and learning environment provided throughout the successful completion of the project "Curelio - Machine Learning Based Cashless Health Insurance Eligibility Prediction System".

This project allowed us to apply our knowledge of Machine Learning, Data Science, and Web Development by building an end-to-end application using Python, Flask, and Machine Learning techniques.

We would also like to acknowledge the resources, tools, and technologies that helped us improve our technical skills and complete this project successfully.

Working on this project enhanced our understanding of data analysis, model development, deployment, and creating AI-based solutions for real-world healthcare challenges.

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INTRODUCTION

Curelio is a Machine Learning based Cashless Health Insurance Eligibility Prediction System developed as part of an internship

Healthcare insurance verification is an important process where patients, hospitals, and insurance providers need accurate and quick decisions regarding cashless treatment eligibility. Traditional verification methods often depend on manual checking, which may increase processing time and reduce efficiency.

Curelio aims to simplify this process by using Machine Learning techniques to analyze important information such as insurance provider, hospital details, policy type, and treatment information. Based on these factors, the system predicts whether the patient is eligible for cashless insurance services.

The project includes data preprocessing, exploratory data analysis, machine learning model training, model evaluation, and integration with a Flask-based web application. Multiple algorithms are compared, and the best-performing model is selected for prediction.

By combining Artificial Intelligence, Machine Learning, and Web Technology, Curelio demonstrates how intelligent systems can support faster and smarter decision-making in the healthcare insurance sector.

PROBLEM STATEMENT

In the healthcare industry, cashless health insurance verification plays an important role in providing quick medical services to patients. However, the traditional verification process often involves manual checking between hospitals and insurance providers, which can lead to delays and increased waiting time.

Patients may face difficulties in identifying whether their insurance policy is eligible for cashless treatment at a particular hospital. Manual processing can also increase the workload for healthcare administrators and may reduce the efficiency of decision-making.

The main problem addressed by this project is the need for a faster and smarter approach to predict cashless health insurance eligibility using available healthcare and insurance-related data.

Curelio aims to solve this challenge by developing a Machine Learning based prediction system that analyzes important factors such as insurance provider, hospital information, policy type, and treatment details to provide an eligibility prediction through an easy-to-use web application.

EXISTING SYSTEM

In the existing healthcare insurance process, cashless eligibility verification is mostly performed through manual communication between patients, hospitals, and insurance providers.

When a patient requests cashless treatment, the hospital verifies the insurance policy details and sends the required information to the insurance provider. The approval decision depends on manual document verification, policy checking, and communication between different departments.

This traditional approach can require more processing time and may cause delays during important medical situations. It also increases the workload for hospital administrators and insurance teams.

The absence of an automated prediction system makes it difficult to provide quick information about cashless eligibility. Patients may need to wait until the verification process is completed before receiving confirmation.

Therefore, there is a need for an intelligent system that can analyze insurance-related data and assist in providing faster eligibility predictions.

Limitations of existing System:

- Time-consuming manual verification
- Increased administrative workload
- Delayed eligibility confirmation
- Limited automation in decision support

PROPOSED SYSTEM

The proposed system, Curelio, is a Machine Learning based Cashless Health Insurance Eligibility Prediction System designed to provide a faster and more efficient approach for insurance verification.

Curelio uses machine learning techniques to analyze healthcare and insurance-related information such as insurance provider, hospital details, policy type, and treatment information. Based on these features, the system predicts whether cashless health insurance eligibility is available or not.

The system includes data preprocessing, feature engineering, model training, and performance evaluation. Multiple machine learning algorithms are implemented and compared, and the most accurate model is selected for integration into the application.

A Flask-based web application is developed to provide a simple and interactive user interface where users can enter the required details and receive prediction results instantly.

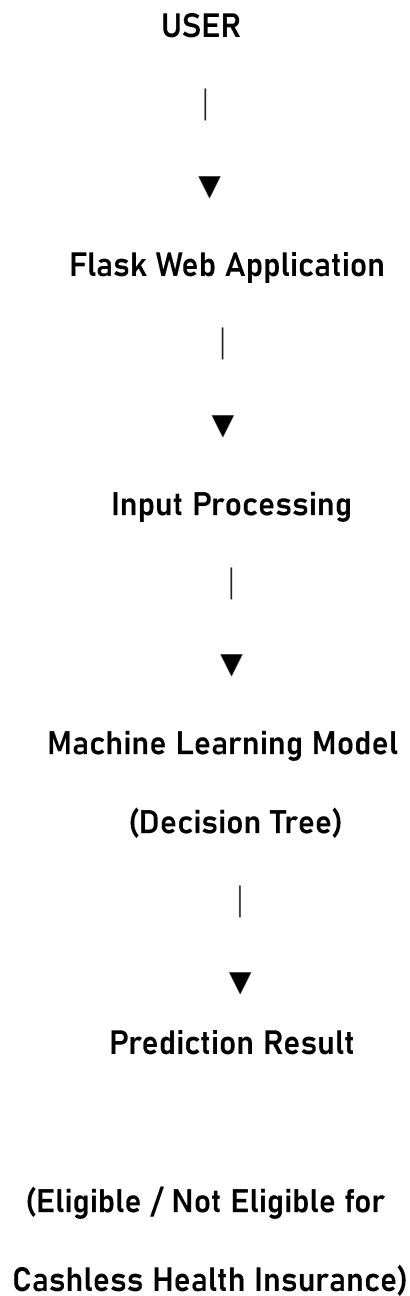
By automating the eligibility prediction process, Curelio helps in reducing manual effort, improving response time, and demonstrating the application of Artificial Intelligence in healthcare insurance decision support.

Advantages of Proposed System:

- Faster eligibility prediction
- Reduces manual verification effort
- User-friendly web interface
- Machine Learning based decision support

Improves efficiency in insurance processing

SYSTEM ARCHITECTURE



EXPLANATION:

The system architecture of Curelio represents the complete workflow of the Cashless Health Insurance Eligibility Prediction System.

The user interacts with the Flask-based web application by providing required healthcare and insurance details such as insurance provider, hospital information, policy type, and treatment details.

The entered information is processed and converted into a suitable format using preprocessing techniques. The processed data is then passed to the trained Machine Learning model.

The Decision Tree model analyzes the given input based on previously trained data patterns and generates the prediction result.

Finally, the predicted output is displayed to the user through the web interface, providing a simple and efficient eligibility checking experience.

TECHNOLOGY STACK

The Curelio system is developed using modern technologies from Machine Learning, Data Science, and Web Development. Each technology plays an important role in building an end-to-end intelligent prediction system, from data analysis and model training to web application deployment.

Programming Language

Python

Python is used as the core programming language for implementing machine learning workflows, data preprocessing, model training, and backend application development

Data Processing & Analysis

Pandas

Used for dataset handling, data cleaning, preprocessing, and organizing healthcare insurance data for machine learning operations.

NumPy

Used for performing numerical computations and efficient data operations during the model development process.

Data Visualization

Matplotlib & Seaborn

Used for Exploratory Data Analysis (EDA) by creating visual representations of data patterns, feature relationships, and insights.

Machine Learning Framework

Scikit-learn

Used for implementing machine learning algorithms, data splitting, model training, accuracy evaluation, and prediction generation.

Machine Learning Algorithms

- Decision Tree Classifier
- Random Forest Classifier
- Logistic Regression

Multiple algorithms were trained and compared based on performance, and the best performing model was selected for integration.

Model Storage

Joblib is Used to save and load the trained machine learning model and label encoders for real-time predictions in the web application

Backend Development

Flask

Flask is used to develop the backend server that connects the trained machine learning model with the user interface and handles prediction requests.

Frontend Development

HTML & CSS

Used to design a simple and interactive user interface where users can enter details and view prediction results.

Development & Deployment Tools

Visual Studio Code

Used as the primary code editor for application development.

GitHub

Used for version control, project management, and maintaining the source code repository.

Render

Used for deploying the Flask web application and making the system accessible online.

DATASET DESCRIPTION

The dataset used in the Curelio project contains healthcare and insurance-related information required for predicting cashless health insurance eligibility.

The data includes details about insurance providers, hospitals, policy information, treatment categories, and eligibility status. This dataset is used to train machine learning algorithms to identify patterns and generate accurate predictions.

Before model development, the dataset undergoes preprocessing steps such as data cleaning, handling missing values, encoding categorical features, and preparing the data for machine learning algorithms.

Feature	Description
Insurance provider	Represents the patient's insurance company
city	Location information of the hospital
Hospital Name	Hospital selected for cashless treatment
Policy type	Type of insurance policy available
Treatment	Medical treatment category
Eligibility Status	Target output showing cashless eligibility

Dataset Usage:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Feature transformation using encoding techniques
- Training and testing machine learning models
- Predicting cashless insurance eligibility

DATA PREPROCESSING & FEATURE ENGINEERING

Data preprocessing is an important step in the machine learning workflow that transforms raw data into a suitable format for model training.

In the Curelio system, preprocessing techniques are applied to improve dataset quality, remove inconsistencies, and prepare healthcare insurance data for accurate prediction.

Data Cleaning

The dataset is analyzed and cleaned by identifying missing values, incorrect entries, and duplicate records. Cleaning helps improve the quality and reliability of the machine learning model.

Feature Selection

Important features related to insurance eligibility prediction are selected from the dataset.

Selected Features:

- Insurance Provider
- City

- Hospital Name
- Policy Type
- Treatment Information

Label Encoding

Machine learning models cannot directly process categorical text data. Therefore, categorical features are converted into numerical format using Label Encoding techniques.

Examples:

Insurance Provider → Numerical Values

Hospital Name → Numerical Values

Policy Type → Numerical Values

Train-Test Splitting

The processed dataset is divided into training and testing data. Training data is used to teach the machine learning algorithm, while testing data is used to evaluate model performance on new data.

Model Preparation

After completing preprocessing and feature engineering, the prepared dataset is used for training different machine learning algorithms and comparing their performance.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is performed to understand the structure, patterns, and relationships present in the healthcare insurance dataset.

EDA helps in identifying important insights from the data before building machine learning models. It provides a better understanding of feature distributions, relationships between variables, and data quality.

Objectives of EDA

- Understand dataset structure and characteristics
- Analyze relationships between different features
- Identify patterns affecting insurance eligibility
- Detect missing values and data inconsistencies
- Support better feature selection for model training

Visualization Techniques

Count Plot:

Used to analyze the distribution of categorical features such as insurance providers, hospitals, policy types, and eligibility status.

Bar Chart:

Used to compare different categories and understand feature relationships.

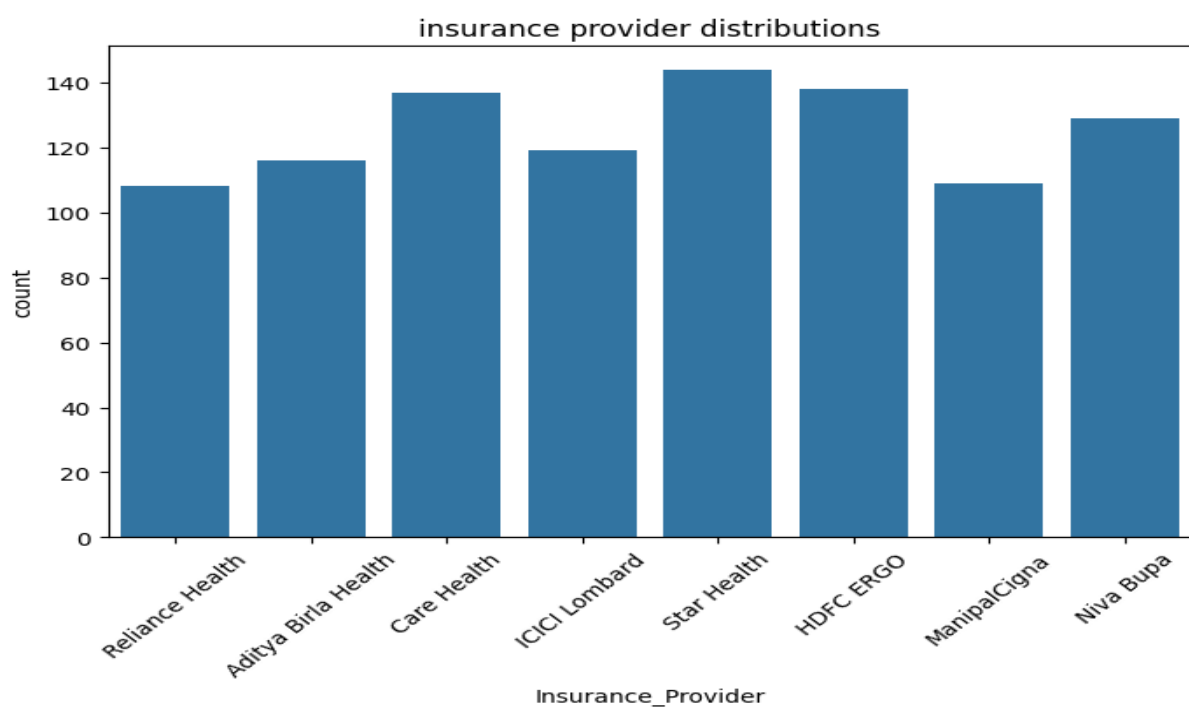
Correlation Analysis:

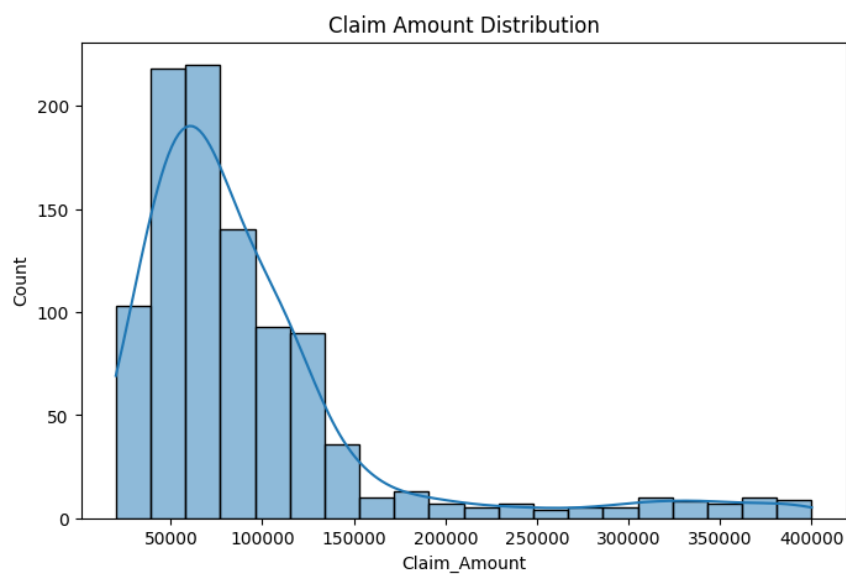
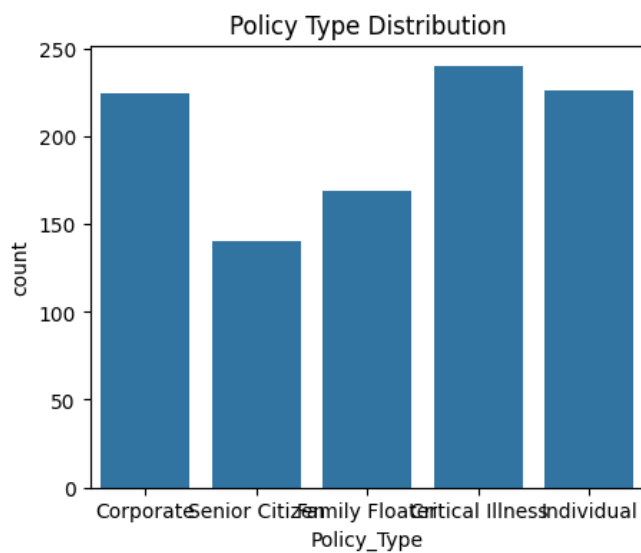
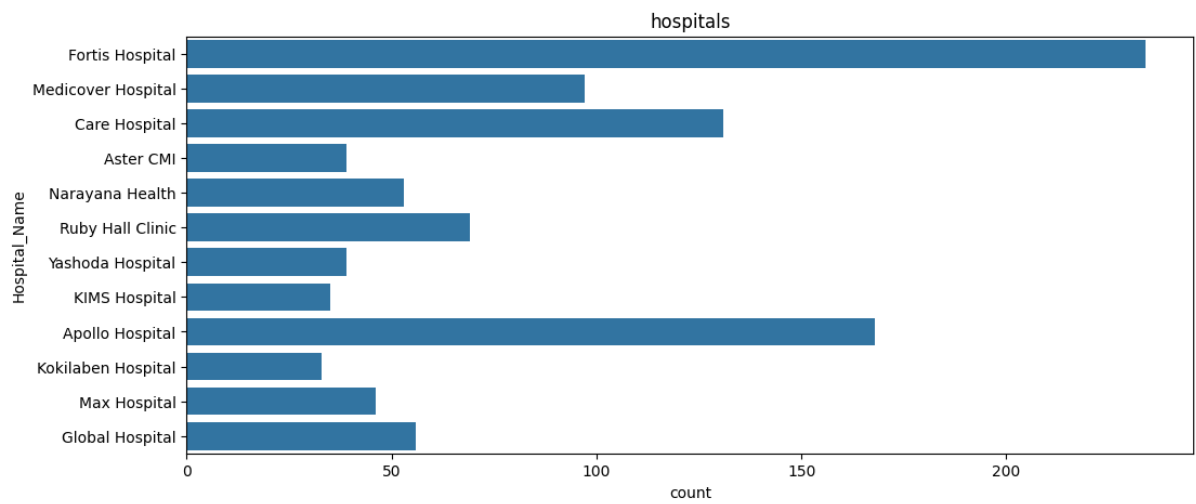
Used to understand relationships between numerical features in the dataset.

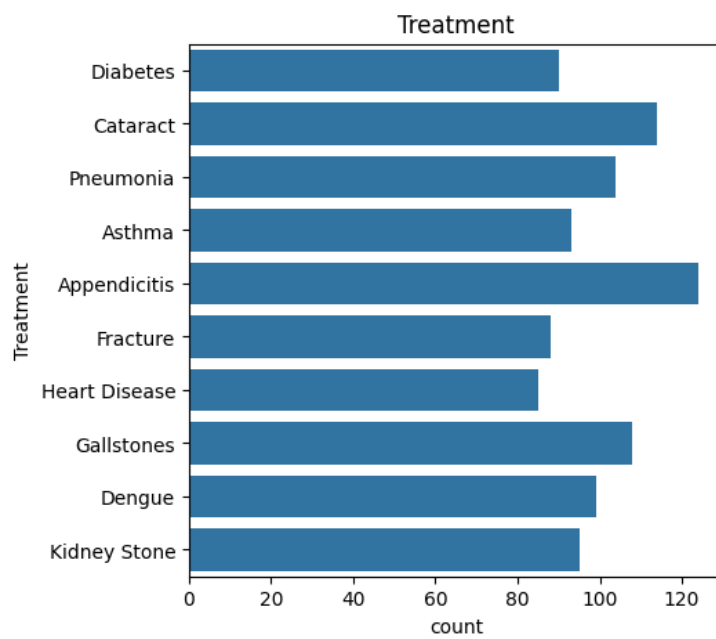
Insights from EDA:

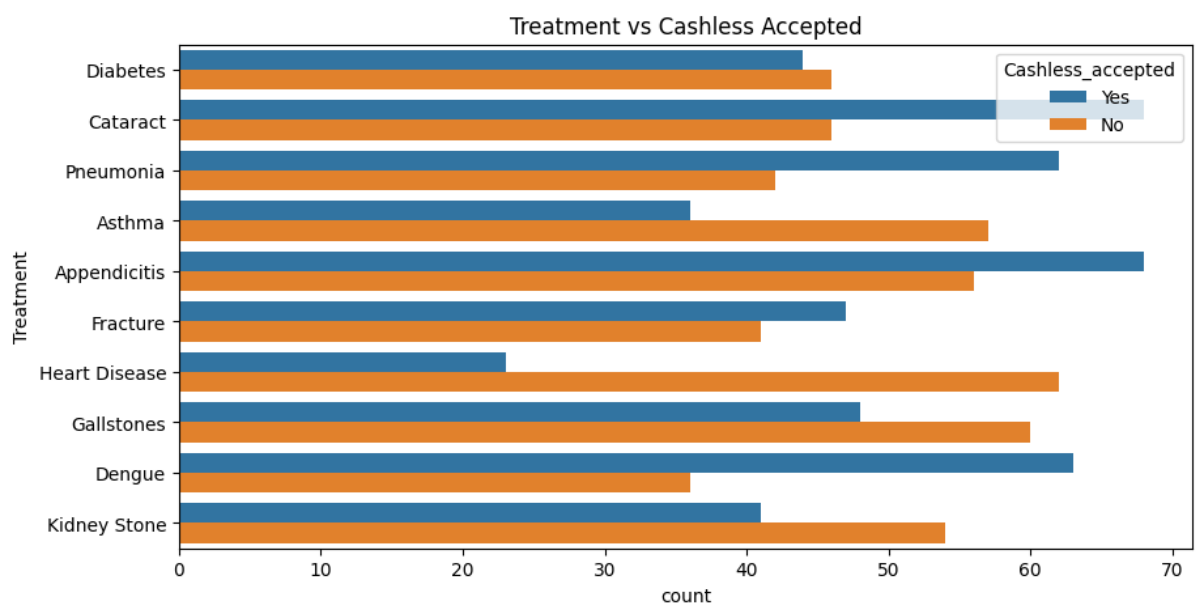
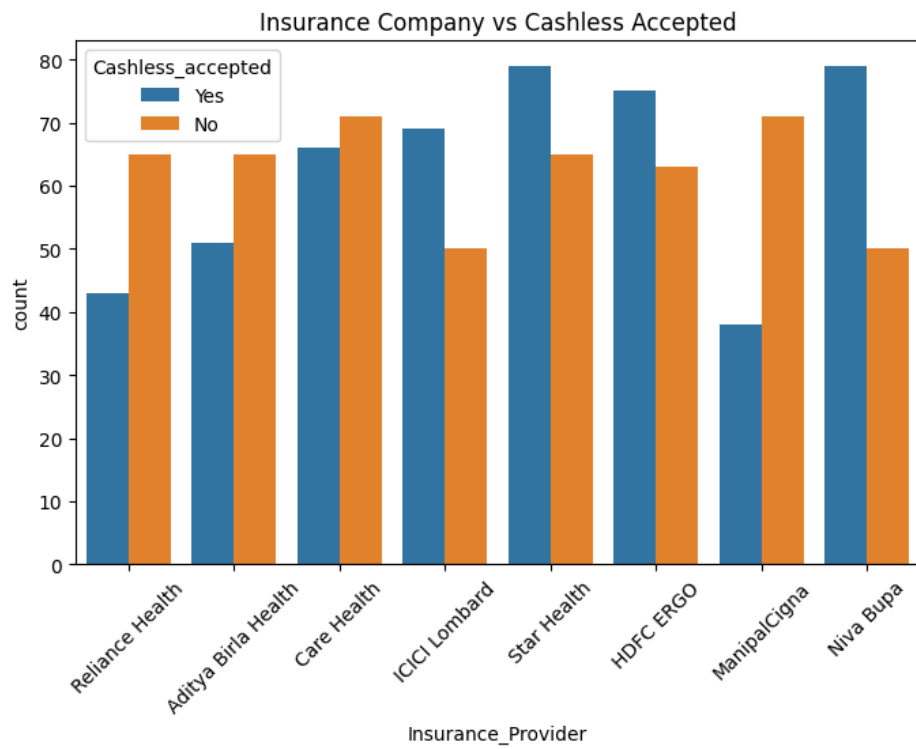
The analysis of the dataset helped in understanding important factors that influence cashless health insurance eligibility predictions.

The insights gained from visualization supported data preprocessing, feature selection, and machine learning model development









MACHINE LEARNING MODELS

Machine Learning models are implemented in the Curelio system to predict cashless health insurance eligibility based on healthcare and insurance-related features.

After completing data preprocessing and feature engineering, multiple classification algorithms are trained and evaluated. The performance of different models is compared, and the most suitable model is selected for integration with the Flask web application.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of a target class based on the relationship between input features.

In Curelio, Logistic Regression is used as a baseline model to analyze insurance eligibility classification performance.

Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

It analyzes different patterns in the dataset and generates predictions based on the combined output of multiple trees.

Decision Tree Classifier

Decision Tree is a supervised machine learning algorithm that makes predictions by creating decision rules from the given dataset features.

It divides data into different decision paths and provides clear classification results.

After comparing model performances, the Decision Tree model is selected for the Curelio system because of its accuracy, simplicity, and effective performance on the dataset.

Model Storage

The trained machine learning model and label encoders are saved using Joblib files. These saved files are loaded into the Flask application to perform real-time predictions from user inputs.

SELECTED MODEL: Decision Tree Classifier

MODEL EVALUATION & COMPARISON

Model evaluation is performed to analyze the performance of different machine learning algorithms used in the Curelio system.

After training multiple models, each algorithm is tested using unseen data to measure prediction performance. The evaluation helps in selecting the most effective model for cashless health insurance eligibility prediction.

Evaluation Metrics

Accuracy Score:

Accuracy measures how correctly the machine learning model predicts the eligibility results compared to the actual values.

A higher accuracy value indicates better prediction performance.

Model Performance Comparison

Algorithm	Accuracy
Logistic Regression	70.0 %
Random Forest Classifier	89.5 %
Decision Tree Classifier	90.5 %

Selected Model

Based on the evaluation results, the Decision Tree Classifier is selected as the final prediction model for the Curelio system.

The Decision Tree model achieved better performance compared to other implemented algorithms and provided efficient classification results for insurance eligibility prediction.

Reason for Selecting Decision Tree

- Better prediction accuracy
- Suitable for classification problems
- Easy to understand decision-making process
- Efficient performance with categorical features
- Simple integration with Flask application

SYSTEM IMPLEMENTATION

The Curelio system is implemented as an end-to-end Machine Learning web application that combines data science, model development, and web technologies.

After completing data preprocessing, model training, and evaluation, the selected Decision Tree model is integrated with a Flask-based web application to provide real-time cashless health insurance eligibility predictions.

Implementation Workflow

1. Dataset Collection

Healthcare insurance data is collected and prepared for machine learning model development.

2. Data Preprocessing

The dataset is cleaned, transformed, and encoded to convert raw data into a suitable format for model training.

3. Model Development

Multiple machine learning algorithms are trained and evaluated. The best-performing Decision Tree model is selected.

4. Model Integration

The trained model and label encoders are stored using Joblib and connected with the Flask application.

5. User Prediction

Users enter required details through the web interface, and the system generates the eligibility prediction result.

Project Structure

Curelio-ML-Prediction-System

- ├── static
 - | └── CSS files
- ├── templates
 - | └── HTML files
- ├── app.py
 - Flask application
- ├── decision_tree_model.pkl
 - Trained ML model
- ├── label_encoders.pkl
 - Encoders for input conversion
- ├── requirements.txt
 - Required dependencies
- ├── Procfile
 - Deployment configuration

Web Application

The user interface is designed using HTML and CSS, while Flask manages backend operations. User inputs are processed, converted into model format, and passed to the trained model for prediction.

The final prediction result is displayed through the web application.

RESULTS AND OUTPUT

The Curelio system successfully predicts cashless health insurance eligibility by analyzing user-provided healthcare and insurance information.

After completing model training and evaluation, the selected Decision Tree Classifier model is integrated with the Flask web application to generate real-time prediction results.

The developed system provides a simple interface where users can enter the required details and receive eligibility predictions quickly.

Model Results

The implemented machine learning models were evaluated based on their prediction accuracy. Among the tested algorithms, the Decision Tree Classifier provided the best performance and was selected as the final model.

Model Accuracy

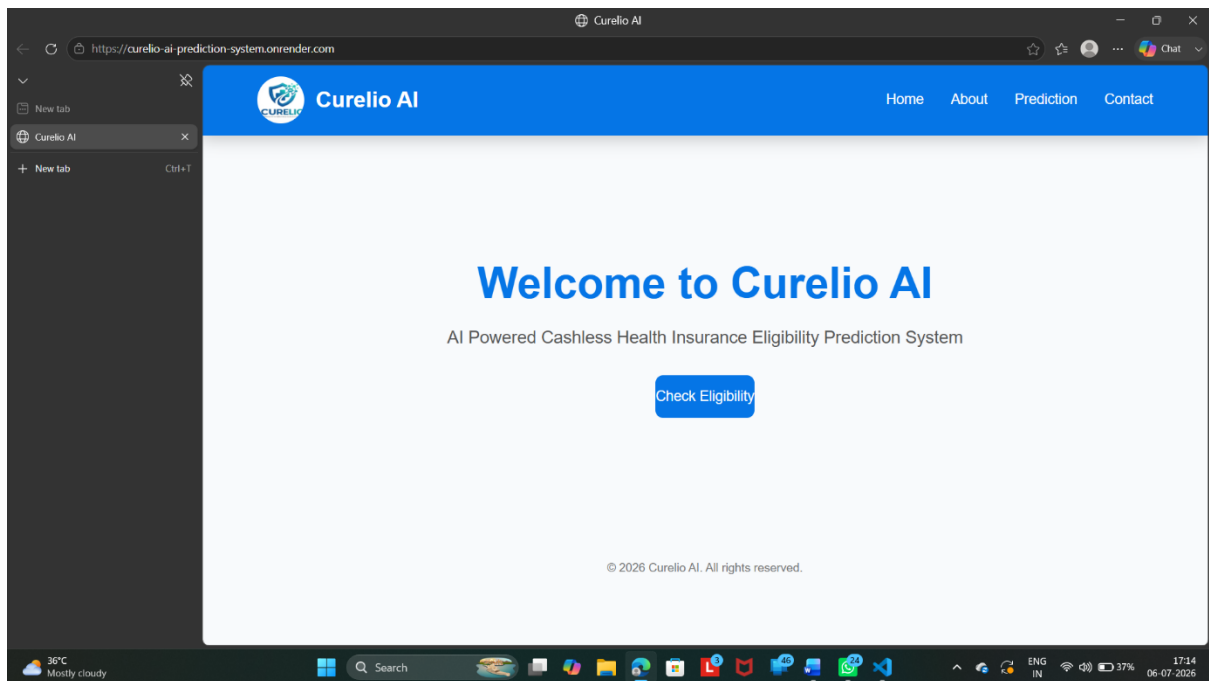
Logistic Regression	70.0 %
Random Forest	89.5 %
Decision Tree Your	90.5 %

Application Output

The Flask web application allows users to:

- Enter insurance and healthcare details
- Process information through the trained ML model
- View cashless eligibility prediction results instantly

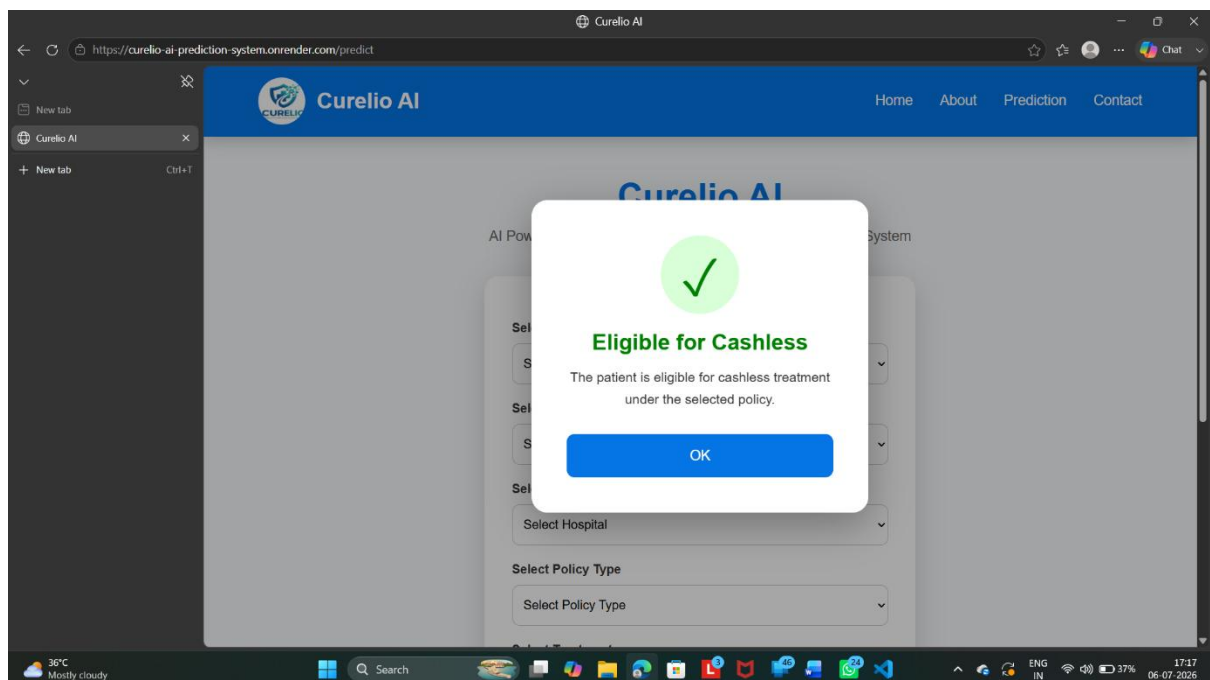
Curelio Web Application Interface



User Input Form

A screenshot of a web browser displaying the 'Curelio AI' prediction system interface. The browser's address bar shows the URL 'https://curelio-ai-prediction-system.onrender.com/predict'. The page features a central white form with several dropdown menus and a text input field. The form fields are labeled: 'Select Insurance Provider', 'Select City', 'Select Hospital', 'Select Policy Type', 'Select Treatment', and 'Claim Amount (₹)'. Each dropdown menu has a placeholder text 'Select [Field Name]'. The 'Claim Amount (₹)' field is a text input with the placeholder 'Enter Claim Amount'. A blue 'SUBMIT' button is located at the bottom of the form. The browser's taskbar at the bottom shows the system time as 17:15 on 06-07-2026, along with various system icons and a weather widget indicating 36°C and 'Mostly cloudy'.

Cashless Eligibility Prediction Result



FUTURE ENHANCEMENTS

The current version of Curelio successfully demonstrates the use of Machine Learning for cashless health insurance eligibility prediction. Future improvements can enhance the system with more advanced features, real-time integration, and better user experience.

Future Scope

1. Real-Time Insurance Integration

The system can be connected with real hospital and insurance provider databases to provide live eligibility verification and updated policy information.

2. Advanced Machine Learning Models

Future versions can include advanced algorithms and deep learning techniques to improve prediction accuracy and handle larger healthcare datasets.

3. Mobile Application Development

A dedicated mobile application can be developed to make the system more accessible for patients, hospitals, and insurance users.

4. User Account Management

Secure login systems can be added for patients and hospitals to manage insurance details and prediction history.

5. Enhanced Data Security

Security techniques can be implemented to protect sensitive healthcare and insurance-related information.

6. AI-Based Recommendation System

The system can be improved to recommend suitable hospitals and insurance services based on user requirements.

CONCLUSION

Curelio - Machine Learning Based Cashless Health Insurance Eligibility



Prediction System demonstrates the application of Artificial Intelligence and Machine Learning techniques in the healthcare insurance domain.

The project focuses on reducing the complexity of traditional insurance verification by developing an intelligent prediction system that analyzes healthcare and insurance-related information to determine cashless eligibility.

During the development process, data preprocessing, exploratory data analysis, feature engineering, and machine learning model implementation were successfully performed. Multiple classification algorithms were trained and compared, and the best-performing model was integrated into the Flask web application.

The system provides a simple and user-friendly interface where users can enter required details and receive quick prediction results. This approach helps demonstrate how machine learning can support faster decision-making and improve efficiency in healthcare-related processes.

Overall, Curelio represents a practical implementation of data science, machine learning, and web technologies to build an AI-based solution for real-world healthcare insurance challenges.